

reached a stalemate. Games have gotten a lot prettier, and most of us swear we could smell the ocean in *Far Cry*; but then you sneak up behind a couple of baddies, shoot one, his friend somehow doesn't hear the shot you fired from three feet away—from a shotgun, no less—and keeps on walking his predefined sentry line, and you are rudely awakened and brought back to reality. How many times have you shook your head in disbelief and disgust at in-game physics and AI? Even the prettiest-looking games have terrible physics.

Of course, of late, game developers have begun adding some wonderfully realistic physics computation into their engines—and what better example than *Half-life 2*'s Gravity Gun to exemplify these? But who are we kidding? All of us know the difference between a game setting and actual video footage—and it's not the characters or backgrounds that let the cat out of the bag! Have you ever seen an oil spill in a game? It looks more like black coloured water than oil. What about when you're climbing a hill (in a game, of course), the way the rocks tumble down the side of the hill you're climbing just looks fake! The same applies for the flow of water in a stream; ditto for the way a glass window shatters when you shoot a bullet through it, or the lack of any mark or bullet holes in the wall you just emptied a clip into. To get into some morbid details, it's also unrealistic when you shoot a guy in the foot two or three times and kill him—makes us wonder if he died in relief after finally getting rid of that nagging pain his ingrown toenail has caused him for years!

OK, It's Unrealistic, But So What?

Now that we've made our point about the idiotic physics in most games, let's see what can be done about it! Those of us who have had the pleasure of playing games with great physics engines, such as *Half-life 2*, will vouch for the fact that more realistic physics is already possible—it's just a matter of creating or coding the game engine that way! So what's stopping the developers?

There's the minimum system requirement problem: developers just cannot make a game that only runs on four computers in the world—that would be a rather idiotic business decision. So, games are developed so they can be played on most mid-range to high-end modern computers. This means that even though you have a dual-core 4 GHz CPU and 4 GB of RAM, you're playing a game that was also developed with your neighbour in mind, who has a mere 1 GHz CPU and 256 MB of RAM. Sure the game will look prettier on your 7800 GTx than on his TNT2, but the physics will remain the same for both. After all, the game was probably designed to be able to calculate its physics on an 800 MHz CPU!

That's Not Fair!

Sure, but how can you change the fact that game developers actually have companies to run, and will always program with minimum system requirements in mind? Change the way PCs are built, of course...

And that's exactly what American company

A lot of developers have already begun developing games that interface with Ageia's technology to offer much more realistic physics engines

Ageia Inc. plan to do!

Ageia is a fabless semiconductor company, and is the first to develop and sell a PPU (Physics Processing Unit), called PhysX, to the general public. They have developed the hardware and software, but do not manufacture the PPUs themselves—they leave the manufacturing and selling to companies such as Asus, Alienware, Dell and Falcon Northwest.

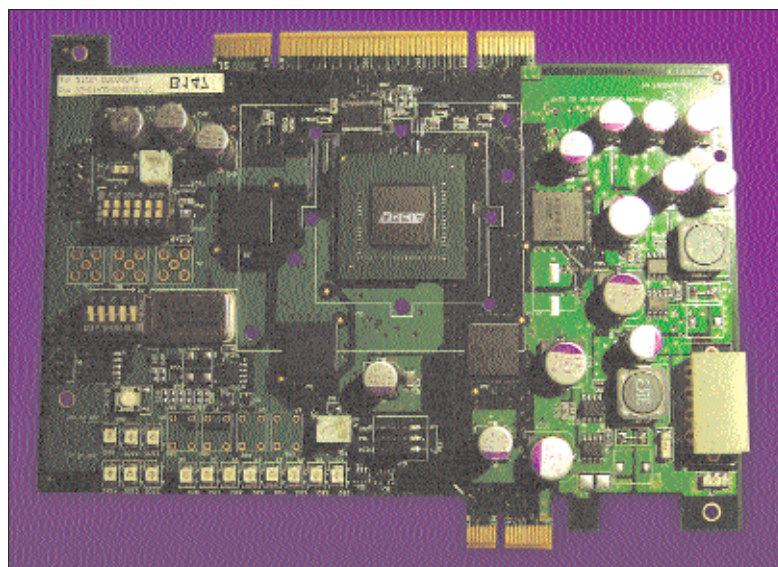
This news is not new to the technology world: Ageia have been showcasing their designs and PPU for some time now. In fact, a lot of developers have already begun developing games that interface with Ageia's technology to offer much more realistic physics engines. You can already go out and buy the Ageia PhysX card—provided you're in the US, of course—as well as a few games that have already been tied into Ageia's technology. A few examples of already-released games are *Rise of Nations: Rise of Legends* and *Tom Clancy's Ghost Recon Advanced Warfighter*. The most noteworthy "soon-to-be-released" game that will showcase the PhysX technology is *Unreal Tournament 2007*.

So We'll All Have PPUs Now As Well?

If you want to have the very best in terms of gaming hardware, yes! Of course, as with other gaming hardware, PhysX will not be cheap. We're not very sure about the prices right now, but the guesstimates say between \$200 and \$400 per card. That's about another 15 thousand rupees you'll have to spend on your gaming rig!

For the first time since graphics and processing computation were separated into the GPU and CPU respectively, the hardware business is in for a complete revamp of sorts. Though PhysX cards will only be available as PCI or PCIe cards initially, motherboard manufacturers, such as Asus, are already looking at integrating PhysX into their boards. Be prepared to see much better-equipped onboard graphics solutions from major motherboard manufacturers soon.

Of course, this innovation will find its way into the consoles soon enough, but it's the PC



Here's the Ageia PhysX board that promises to revolutionise gaming for us all